

Dataflow rounds depend on lattice height

Disproof of Conjecture 00000003689

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The fixed dataflow problem. Use the directed control-flow graph with one vertex v and one self-loop $v \rightarrow v$. Its node count and loop depth are both one. Take the complete information lattice

$$L = \mathbb{N} \cup \{\infty\}, \quad 0 < 1 < 2 < \dots < \infty.$$

The monotone transfer function is $T(a) = a + 1$ for finite a and $T(\infty) = \infty$. Since there is only one incoming edge, the global dataflow update of the state s is $F(s)(v) = T(s(v))$. This is a genuine monotone lattice fixed-point iteration on the stated graph. Start from the bottom state $s_0(v) = 0$.

No finite convergence. Induction gives $s_r(v) = r$ for every finite round r . Hence $F(s_r)(v) = r + 1 \neq r = s_r(v)$: the iteration never stabilizes after any finite number of rounds. The top state is fixed, and is in fact the unique fixed point, since no finite natural number equals its successor. The supremum of the finite orbit is the top state, which is reached only at the limit stage. Thus even one fixed complete lattice and one fixed one-node graph refute any proposed finite convergence count linear in the number of nodes. The conjecture states no finite-height or ascending-chain condition.

Finite truncations and the missing parameter. Even if a finite lattice were intended, no uniform bound depending only on node count and loop depth can hold. On the same graph take

$$L_m = \{0, \dots, m\}, \quad T_m(a) = \min(a + 1, m).$$

This is an actual finite chain lattice and a monotone transfer. Induction yields $s_r^{(m)}(v) = \min(r, m)$, so

$$F_m(s_r^{(m)}) = s_r^{(m)} \iff r \geq m.$$

The first stable state therefore occurs at round m , arbitrarily large with node count and loop depth still one. For any proposed uniform coefficient C , choose $m = C + 1$; no round $r \leq C \cdot 1$ is stable. Counting an extra round to detect stability only increases this cost.

Scope of the disproof. For a fixed finite lattice, a node-count bound whose constant depends on lattice height can be valid. The finite family above specifically disproves a uniform node/depth-only bound. The fixed infinite-height complete-lattice example removes that constant ambiguity altogether, because there is no finite stabilization. A finite-height hypothesis and an explicit height parameter would be substantive additions to the original statement.

Formal certificate. Lean constructs the actual one-node self-loop quiver, the complete extended-natural information lattice, monotone transfer, global update and its function iterates from bottom. It proves the exact orbit, unique fixed point and absence of every finite stable round. It additionally constructs the ordered finite domains, calculates their actual iterates and first stabilization threshold, and rules out every uniform node-only coefficient. Final axiom audits use only standard Lean and Mathlib axioms.